



IUBAT School of Computer Science and Engineering
Department of Computer Science and Engineering

Assignment of Summer Semester 2026

Program: Bachelor of Computer Science and Engineering (BCSE)

Course Instructor: Md. Asif Hossain

Course Code: CSC 439

Course Title: Visual Programming

Section: C & D

Full Marks: 100

Submission Date: 05 – 09 – 2026

CO	1	2	3
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PO	a	b
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CO3 - PO2 - C4 – 100 Marks

Scenario

A university currently receives student requests for services such as **ID card replacement, transcript requests, and certificate requests** through manual forms and email. This makes it difficult for students to track their requests and for university staff to manage them efficiently.

The university wants a simple **Student Service Request System** developed using **ASP.NET MVC**.

The system should allow students to submit a service request and check its status. A staff member should be able to view submitted requests and update their status.

However, the university has two important concerns:

- The system should be **simple and easy to use** for students.
- Only authorized staff should be able to access and modify student requests.

You are required to **analyze the problem, propose a suitable solution, and develop a functional ASP.NET MVC application.**

Tasks

Part A — Problem Analysis

Analyze the given problem and:

1. Identify the main **users/stakeholders** of the system.
2. Identify the major **functional requirements** of the system.
3. Identify the major **non-functional requirements**.
4. Identify the conflict between **ease of use and access control/security**.
5. Explain how this conflict should influence your system design.



Part B — System Design

Based on your analysis, design the proposed solution.

Your design should include:

- The main **Models** required.
- The main **Controllers** required.
- The main **Views** required.
- A simple database structure.
- A basic diagram showing how the **User** → **View** → **Controller** → **Model/Database** components interact.

You should briefly justify why your proposed structure is appropriate for the problem.

Part C — ASP.NET MVC Implementation

Develop a functional prototype with the following minimum features:

Student

A student should be able to:

- Register/login to the system.
- Submit a service request.
- Select a request type, such as:
 - ID Card Replacement
 - Transcript Request
 - Certificate Request
- Provide a short description of the request.
- View their submitted requests.
- View the status of each request.

Staff

A staff user should be able to:

- Login to the system.
- View submitted service requests.
- View the details of a request.
- Change the request status to:
 - **Pending**
 - **Processing**
 - **Completed**
 - **Rejected**



Requirements

The application must demonstrate appropriate use of:

- ASP.NET MVC
- Models and ViewModels where appropriate
- Controllers and Actions
- Razor Views
- Routing
- HTML Forms
- Server-side validation
- Database connectivity
- CRUD operations
- Appropriate navigation between pages
- Basic role-based access or authorization for student and staff functions

Students may use **Entity Framework** or another database-access approach taught in the course.

Deliverables

Students must submit:

1. **ASP.NET MVC project/source code**
2. **Database/database backup**
3. **Short report with KPA Form Attachment**
4. **Short demonstration of the application**



Assignment Report Format

1. Introduction

Briefly describe:

- Background of the problem
- Problem statement
- Objectives of the proposed system

Recommended length: 1 page

2. Problem Analysis

2.1 Stakeholders and Requirements

Identify the major stakeholders and their key functional and non-functional requirements.

2.2 Conflicting Requirements

Identify **two conflicting requirements** and explain how they influence the system design.

Examples:

- Usability vs. Security
- Simplicity vs. Functionality

Recommended length: 1 page



3. System Design

3.1 Proposed Solution

Briefly explain how the proposed ASP.NET MVC system addresses the identified problem.

3.2 MVC Design

Identify the major:

- Models
- Controllers
- Views

Include a simple **MVC interaction diagram**.

3.3 Database Design

Provide the main database tables and their relationships.

Recommended length: 2 pages

4. Implementation

Briefly describe the implementation using ASP.NET MVC.

Discuss:

- ASP.NET MVC features used
- Database connectivity
- CRUD operations
- Form validation
- Student/staff access control

Include screenshots of the major system interfaces.

Recommended length: 2–3 pages



6. Conclusion

Briefly summarize:

- The problem addressed
- The developed solution
- Major ASP.NET concepts applied
- Key design decisions and findings

Recommended length: 1 page

References

List the main books, documentation, websites, or other resources used.

KPA Form



Expected KPA to be Covered:

- **K3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- **K4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- **P1 Depth of knowledge required:** Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
- **P2 Range of conflicting requirements:** Involve wide-ranging or conflicting technical, engineering and other issues
- **P3 Depth of analysis required:** Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models

Bloom's Taxonomy Levels in Cognitive Domain:

C1: Remember; **C2:** Understand; **C3:** Apply; **C4:** Analyze; **C5:** Evaluate; **C6:** Create

Course Outcomes:

CO1: Explain specific design patterns and different features of visual programming language.

CO2: Apply design patterns and visual programming language features to solve a problem.

CO3: Analyze the use of visual programming features to develop a solution to a real-life problem.

Program Outcomes:

PO1: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.

PO2: Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).